

PROJECT FALCON

Wednesday Presentation Guide

Clear slide-by-slide script · Target duration: 9–11 minutes

Main message: Project FALCON is an affordable, local-first coastal monitoring buoy concept with a working software foundation. The funding request is intended to build, calibrate, and field-test the physical system. Do not claim that its wave prediction or sensors are already field validated.

Before You Present

- Assign one main speaker and one member who will operate the slides and dashboard.
- Speak slowly. Pause after every major idea instead of reading the slide word for word.
- Use the slide as visual evidence; use this guide to explain what the audience should understand.
- If interrupted by a question, answer briefly and return using the transition line.

Simple Component Guide

Easy way to explain the hardware: Do not begin with model numbers. First say the component's simple job, then mention its name. Example: "This sensor measures the tilt and movement of the buoy. It is called the BNO085."

Component	Simple meaning and purpose	Easy line to say
Main float	The round floating body that supports the complete structure and provides stability on the water.	"This is the body that keeps FALCON floating and supports all equipment above the water."
Tower and frame	The strong structure that holds the sensors, solar panels, antennas, and electronics pod at their intended positions.	"The tower keeps the equipment secure and gives the upper sensors a clear position."
Upper electronics pod	The sealed and serviceable box that protects the computers and electronic boards from water and weather.	"This is the dry control box or brain compartment of the buoy."
Wind sensor / SEN-15901	The anemometer measures wind speed, while the wind vane identifies wind direction.	"This tells us how fast the wind is moving and where it is coming from."
GPS module	Provides the buoy's position, movement, heading-related information, and satellite-fix quality.	"GPS tells us where the buoy is and helps us detect if it moves unusually far from its deployment point."
BNO085 IMU	An inertial measurement unit that measures orientation and motion, including roll, pitch, and rotational movement.	"This is the motion sensor. It tells us how the buoy tilts and moves with the waves."

Bar02 pressure sensor	Measures water pressure. After calibration and depth correction, changes in pressure may help estimate water movement and wave characteristics.	"This sensor feels changes in water pressure, which may help us study waves after proper calibration."
Optional DS18B20	A waterproof temperature probe intended for measuring water temperature if it is included in the approved final scope.	"This optional probe measures the temperature of the surrounding water."
ESP32 DevKit V1	The microcontroller that reads sensors repeatedly, performs basic checks, and packages the readings for transmission.	"The ESP32 is the sensor reader. It gathers the raw measurements and sends them in an organized format."
Orange Pi Zero 3	The small Linux computer that stores data, performs heavier calculations, runs the API, hosts the dashboard, and later executes the validated model.	"The Orange Pi is the onboard computer. It saves, processes, and presents the information."
MicroSD or eMMC storage	Stores the operating system, database, dashboard files, sensor history, and test records.	"This is the buoy's local memory, so records can remain available even without internet."
ADS1115	A high-resolution analog-to-digital converter used to read the changing electrical signal from the wind-direction vane more consistently.	"This converts the wind vane's electrical signal into a value the ESP32 can understand."
MCP9808	Measures temperature inside the electronics enclosure to identify overheating risk.	"This checks whether the electronics box is becoming too hot."
Two INA260 monitors	One board measures the battery branch and the other measures the solar branch. They report voltage, current, and power flow.	"These are power meters: one watches the battery and one watches the solar charging."
Solar panels	Convert sunlight into electrical energy for charging the battery and supporting longer operation.	"The solar panels provide renewable energy for the buoy."
MPPT solar charger	Controls energy transfer from the solar panels to the battery and helps charging operate efficiently and safely.	"This is the charging controller between the solar panels and the battery."
12.8 V LiFePO4 battery	Stores solar energy and powers the system when sunlight is weak or unavailable. Final capacity still requires measured load testing.	"The battery stores energy so FALCON can continue operating when there is no sunlight."
Fuse and disconnect	The fuse protects wiring during an electrical fault, while the disconnect allows power to be safely isolated during maintenance.	"These are safety parts that protect the system and let us turn off power before servicing it."
5 V buck converters	Safely reduce the battery voltage to regulated 5 V supplies. Separate branches are planned for the Orange Pi and ESP32 peripherals.	"These lower the battery voltage to the safe voltage required by the computers."

Terminal blocks	Provide organized connection points for power, ground, and signals, making inspection and replacement easier.	“These keep the wire connections organized, labeled, and easier to repair.”
Cable glands and connectors	Secure cables entering the enclosure and help provide strain relief and water protection.	“These protect the cable entry points so wires are not easily pulled or exposed to water.”
Mooring and anchor	Hold the buoy near its approved deployment position while allowing normal movement caused by tide, current, and waves.	“The mooring keeps the buoy in its assigned area, while GPS helps us check abnormal movement.”
Local database	An organized record of measurements, predictions, alerts, and system events stored on the edge computer.	“The database is the digital logbook of everything the buoy records.”
API	A software connection that lets the dashboard request current and historical information from the edge service.	“The API is the messenger between the saved data and the dashboard.”
Dashboard	The human-readable interface that displays current readings, graphs, history, alerts, and system health.	“The dashboard turns technical sensor data into information that people can understand.”

One-Sentence System Explanation

“The sensors observe the buoy and environment; the ESP32 collects their readings; the Orange Pi saves and processes the data; the solar and battery system supplies energy; and the dashboard explains the results to the user.”

Slide-by-Slide Script

Slide 1 — Project FALCON (35–45 seconds)

Goal: Introduce the team, the project, and the purpose of the presentation.

“Good day, everyone. We are the Project FALCON Research Group from Fullbright College. Project FALCON is our proposed smart buoy for observing coastal conditions in one specific location. In simple terms, it is a floating device with sensors, a small computer, solar power, and a dashboard. It will collect information near the deployment site and show the data in an easier form. Today, we will explain the problem, our proposed solution, what already works in software, and what we still need to build and test.”

Transition: “To understand why we designed FALCON, let us first look at the monitoring gap.”

Slide 2 — Why This Matters (50–60 seconds)

Goal: Explain the problem without criticizing PAGASA or claiming that no coastal information exists.

“Why is this project needed? We already have official forecasts, and we respect them. FALCON will not replace PAGASA or any official warning system. Our concern is that a forecast for a large area may not describe the exact condition at a small near-shore site. Professional monitoring equipment can also be expensive and difficult for a school or small local organization to maintain. So our question is simple: can we build a more affordable buoy that provides useful information for one selected pilot site in Palawan?”

Transition: “Our response to this gap is a small monitoring system that can measure, process, store, and explain data locally.”

Slide 3 — Proposed Solution (55–65 seconds)

Point to Measure, Process Locally, Display and Alert, and Research Prediction in that order.

“Our proposed solution is a smart coastal buoy with four basic jobs. First, the sensors measure movement, water pressure, wind, location, temperature, and available power. Second, the ESP32 gathers the sensor readings, while the Orange Pi saves and processes them. The Orange Pi is simply the small onboard computer of the buoy. Third, the dashboard shows the information through numbers, graphs, history, and alerts. Fourth, after collecting enough real data, we will study whether the system can estimate wave height several minutes ahead. For now, the prediction is only a simulated research demonstration and not yet a real operational forecast.”

Transition: “The next image shows where these functions are located on the proposed buoy.”

Slide 4 — Upper Assembly and Sensor Levels (55–65 seconds)

Point from the top sensor deck downward. Do not discuss an underwater pointed section because it is not part of the approved visual.

“This image shows the parts above the water. At the top are the wind sensor, GPS, and antennas because they need open space to work properly. The solar panels are mounted on the tower and provide energy for the system. The large box in the middle is the upper electronics pod. It is a dry and protected box where we plan to place the ESP32, Orange Pi, storage, and other electronic boards. The motion sensor is attached to a rigid part of the structure so it measures the movement of the buoy itself. The round body at the bottom is the main float that keeps the structure above water. Its final balance and waterline will still be tested using the physical prototype.”

Transition: “Inside that structure, the sensors and power system follow this proposed electronics plan.”

Slide 5 — Hardware Concept (60–70 seconds)

Explain function, not every wire. Point first to sensors, then ESP32, Orange Pi, and finally the protected power chain.

“To make this diagram easier, we can divide the components into four groups. The first group is sensing: BNO085 measures buoy movement, Bar02 measures water pressure, GPS gives location, and the weather sensor measures wind. The second group is control: the ESP32 collects all sensor readings, while the Orange Pi saves, processes, and sends the results to the dashboard. The third group is system health: INA260 boards act as power meters for the battery and solar branches, while MCP9808 checks if the electronics box becomes too hot. The fourth group is power: solar panels

produce energy, the MPPT controls battery charging, the battery stores energy, and buck converters provide the safe voltage required by the computers. The fuse and disconnect protect the system and allow safe maintenance. So the simple flow is: sensors measure, ESP32 collects, Orange Pi explains, and the power system keeps everything running. This remains a design guide until we confirm the real components, wiring, and safety calculations.”

Transition: “To make this system practical, the electronics must also be organized for safe maintenance.”

Slide 6 — Maintainable Enclosure (50–60 seconds)

Point to the edge-computing zone, sensor-control zone, terminal area, then cable glands.

“This slide shows how we plan to arrange the electronics inside the upper box. The parts will be mounted on a removable tray, so we can inspect or replace them without removing the whole buoy structure. The Orange Pi side handles computing, while the ESP32 side handles sensor reading. Terminal blocks keep the wires organized instead of leaving many loose connections. Each cable will have a label on both ends, and cables entering the box will pass through waterproof locking glands. We will also separate signal wires from power wires to reduce electrical interference. Before actual deployment, we still need to test heat, waterproofing, wire clearance, and ease of maintenance.”

Transition: “With the hardware organized, this is how information travels through the system.”

Slide 7 — System Workflow (45–55 seconds)

Trace the arrows from sensors to ESP32, Orange Pi, database/API, then dashboard.

“This slide shows the flow of information. First, the sensors take measurements. Next, the ESP32 collects and checks those readings. It sends them to the Orange Pi as telemetry, which simply means organized sensor data. The Orange Pi saves the data, performs calculations, checks possible alerts, and sends the results to the dashboard. The dashboard then presents the information as readable values, graphs, and status messages. We call this local-first because the main processing and storage happen inside the buoy system, even without a continuous internet connection.”

Transition: “We have already implemented the software foundation for this workflow.”

Slide 8 — Current Progress (50–60 seconds)

Emphasize both the completed work and the honest limitation shown in the red/simulation notice.

“What have we finished so far? We already created the responsive dashboard, the basic ESP32 firmware, the data format used by the system, the simulator, the Orange Pi or laptop edge service, the local database, alerts, wiring plans, bill of materials, and project documentation. We also tested whether the firmware, backend service, and dashboard can build and operate in software. However, we want to be honest: the current dashboard data are simulated. They do not yet come from a complete buoy in the sea. Our next stage is to connect the real sensors, calibrate them, build the buoy, and perform supervised field testing.”

Transition: “This clear separation between completed work and future validation is also part of our innovation approach.”

Slide 9 — Innovation (45–55 seconds)

Do not say that the team invented the buoy, AI, or sensors.

“We are not claiming that we invented the buoy, the sensors, or artificial intelligence. These technologies already exist. Our proposed innovation is how we combine them into one affordable and maintainable system for a specific local use. FALCON joins sensors, local processing, solar-power monitoring, data storage, a dashboard, and a future prediction model. It also clearly labels whether information is measured, simulated, estimated, or predicted. Our contribution will be proven only if the complete system is calibrated and tested for one selected coastal site.”

Transition: “The value of that integration depends on who will use it and what specific need it will serve.”

Slide 10 — Beneficiaries (40–50 seconds)

Name possible beneficiaries, then narrow the study to one primary partner.

“Several groups may benefit from this project. A coastal LGU or DRRM office may use the observation history and equipment alerts. Fisherfolk or small-boat users may benefit from easier access to information about one nearby site. Schools and researchers may use the platform and collected dataset for future studies. However, we will not claim that one prototype can immediately serve everyone. We will first choose one partner, one pilot location in Palawan, and one specific information need. If the first pilot is useful, the project can be improved and expanded later.”

Transition: “To reach that verified pilot, we propose a staged development plan.”

Slide 11 — Development Plan (45–55 seconds)

Move through the roadmap from design and procurement to controlled testing and reporting.

“Our estimated development time is eight to twelve months. First, we will finalize the design, exact components, safety requirements, and purchasing plan. Second, we will assemble the electronics and fabricate the buoy enclosure and structure. Third, we will calibrate the sensors and test power, waterproofing, and communication in controlled conditions. After those tests pass, we will conduct a supervised near-shore pilot. We will use the collected data to compare our AI model with a baseline, meaning a simple prediction method used as a reference. Finally, we will report the results, errors, limitations, and recommended improvements.”

Transition: “Completing these stages requires support for more than electronics alone.”

Slide 12 — Preliminary Budget (45–55 seconds)

Point to the categories instead of reading every amount. Call the amount a planning range.

“Our preliminary budget is around seventy-two thousand to one hundred twenty-seven thousand pesos. Why is it a range? Because the exact marine-rated parts, fabrication cost, and supplier prices still need final quotations. The budget includes not only sensors and computers, but also solar power, battery, waterproof enclosure, buoy structure, mooring, safety materials, calibration, field testing, transportation, spare parts, and contingency. In other words, the request covers the complete process of building and testing a safe prototype. Before purchasing, we will submit an updated bill of

materials with actual supplier prices.”

Transition: “With this support, these are the concrete results we intend to deliver.”

Slide 13 — Expected Result and Request (50–60 seconds)

End with the deliverables, specific request, and evidence-focused closing statement.

“If the project receives support, our goal is to produce one complete and documented physical prototype. We will deliver calibrated sensor results, tested solar and battery operation, a working local dashboard, stored coastal data, and supervised pilot-test findings. We will also compare the future AI model with a simple baseline to determine whether AI actually improves the prediction. We are requesting support for the components, fabrication, calibration, field testing, and technical guidance. To be clear, we are not asking you to trust an untested forecast. We are asking for the opportunity to build the system properly and collect evidence that will show whether it works. Thank you, and we are ready for your questions.”

Likely Questions and Simple Answers

Question	Suggested answer
Is the AI prediction already working?	“The interface and forecast pipeline work with simulated data, but the model is not yet field validated. We need calibrated local measurements before training and comparing it with a persistence baseline.”
What makes FALCON different?	“Its contribution is the affordable, local-first, and transparently validated integration of sensing, edge computing, power monitoring, prediction research, and displacement alerts for one defined Palawan use case.”
Will it replace PAGASA?	“No. FALCON is a research and localized observation tool. Official advisories and warnings remain the authority.”
Why use both ESP32 and Orange Pi?	“The ESP32 handles reliable sensor reading. The Orange Pi handles storage, processing, the dashboard, and future model inference. Separating these roles improves reliability and maintainability.”
How will you prove the wave measurements are accurate?	“We will calibrate the sensors, document the buoy and mooring response, and compare synchronized results with a suitable reference such as a pressure logger, staff gauge, video method, or validated wave instrument.”
How will it operate without internet?	“Measurements are processed and stored locally. Internet or long-range communication can be used for remote access, but the core acquisition and archive do not depend on continuous cloud service.”
Why is the budget a range?	“The exact marine-rated components, fabrication method, and suppliers still require final quotation. The range prevents us from presenting an invented exact amount before procurement.”

What happens after funding?	"We finalize the design, purchase verified components, assemble and calibrate the system, perform controlled tests, conduct a supervised pilot, evaluate the model, and submit documented results."
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Final Reminder

If you forget a line, return to this sentence: **"FALCON is a working software concept that needs funded physical development, calibration, and supervised field validation."**